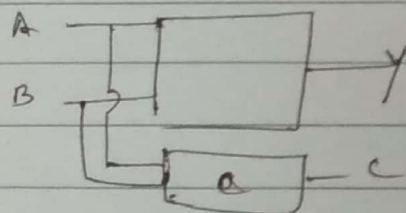
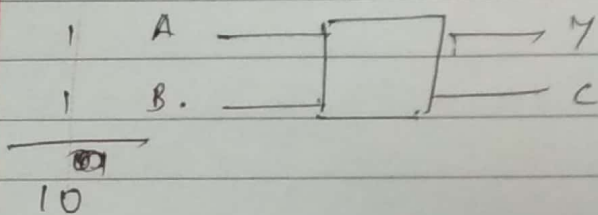
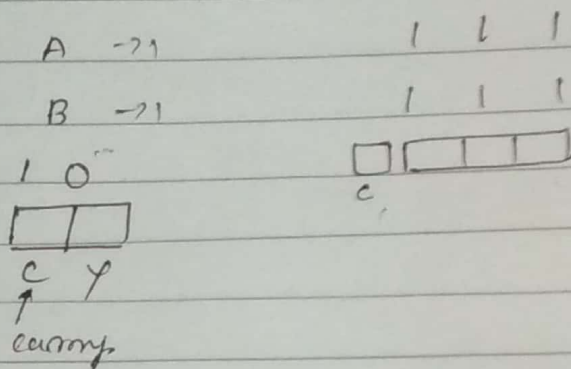
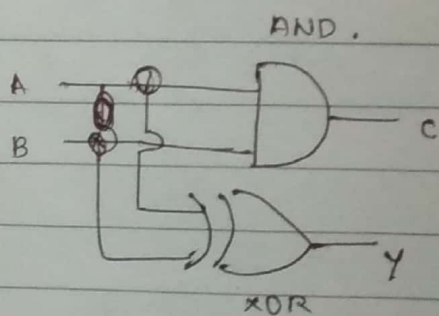


26/02/2018



AND Gate.



carry

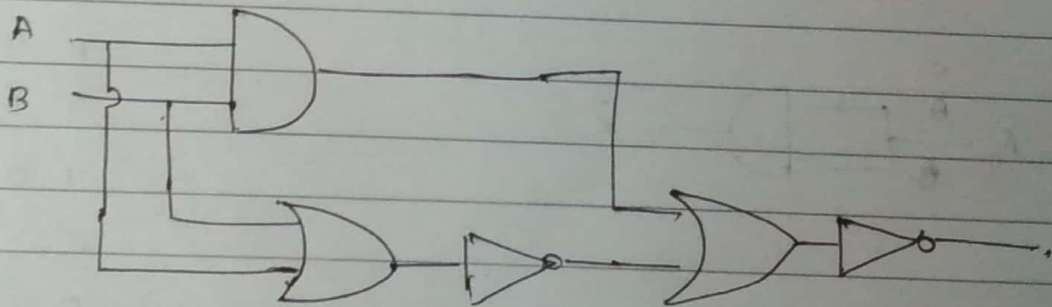
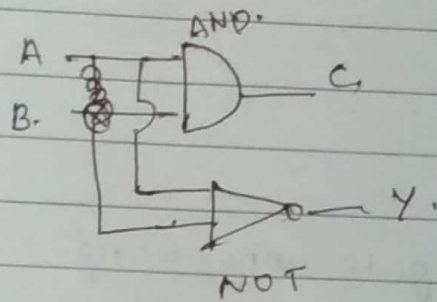
output

A	B	C	Y
1	1	1	0
1	0	0	1
0	1	0	1
0	0	0	0

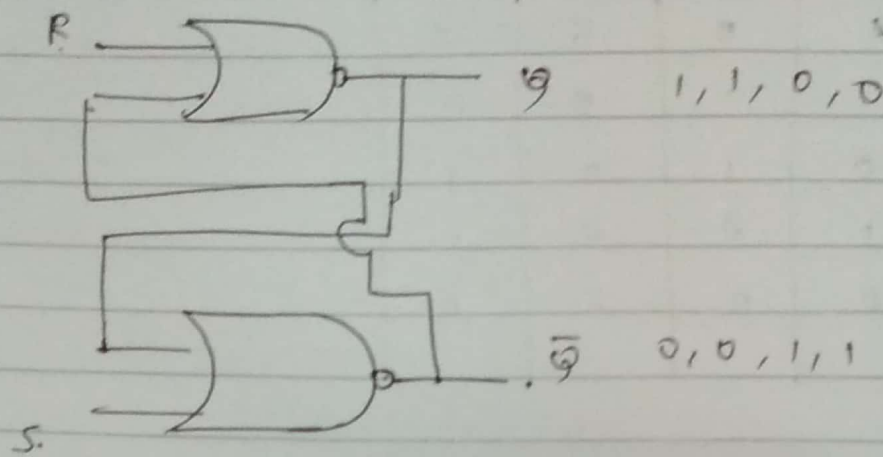
Basic circuit used for
adder ⇒ Adder circuit

Q. Construct an X-OR gate using the basic gates (AND, OR, NOT)

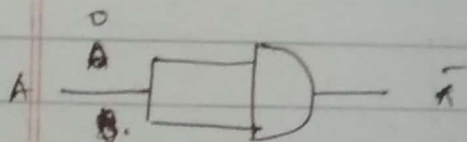
A	B	C	Y
1	1	0	0
0	1	0	1
1	0	0	1
0	0	0	0



Flip Flop circuit



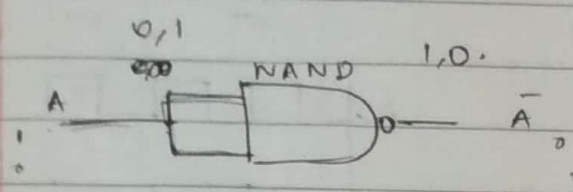
Q. Construct a NAND gate using a combination of AND gates



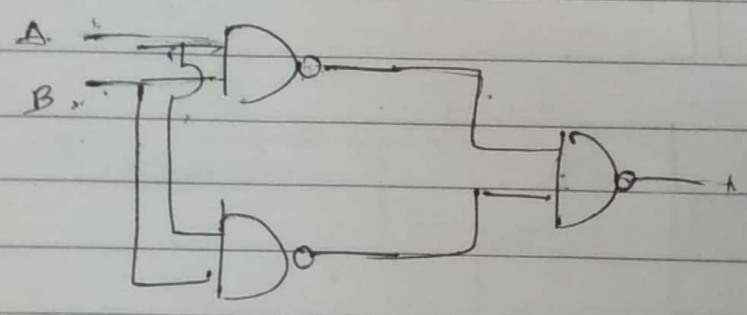
NAND

A	B	Y
1	1	0
1	0	1
0	1	1
0	0	1

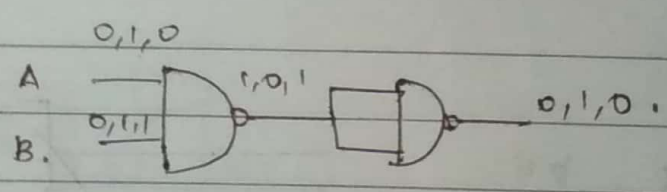
Q2 Using NAND gate construct an AND gate.



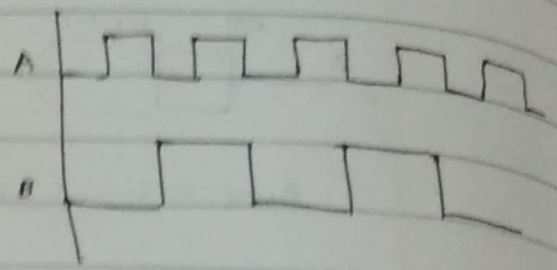
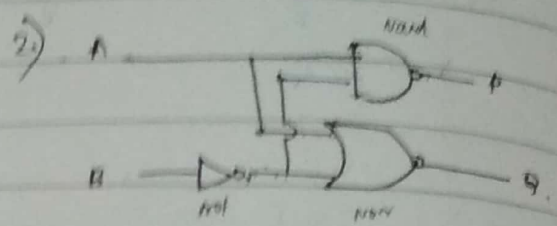
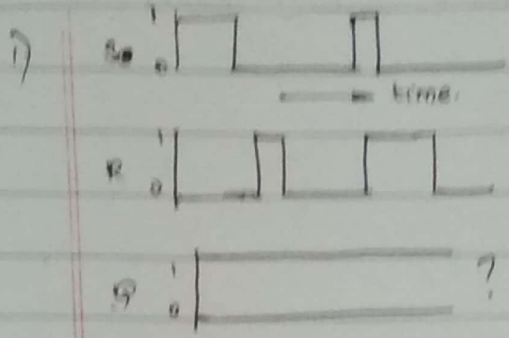
NAND Gate $\Rightarrow$ NOT



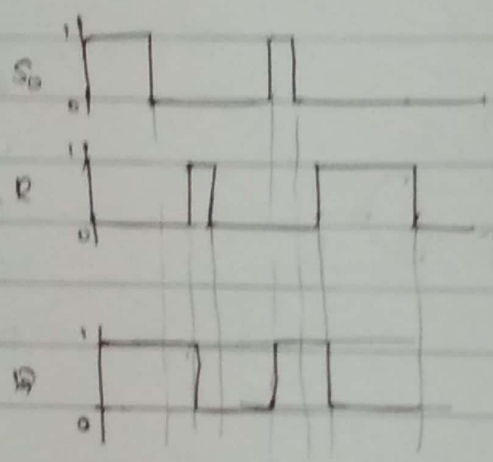
A	B	Y
1	1	1
1	0	0
0	1	0
0	0	1



28/08/2018

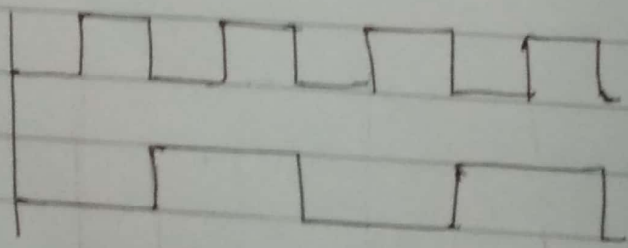
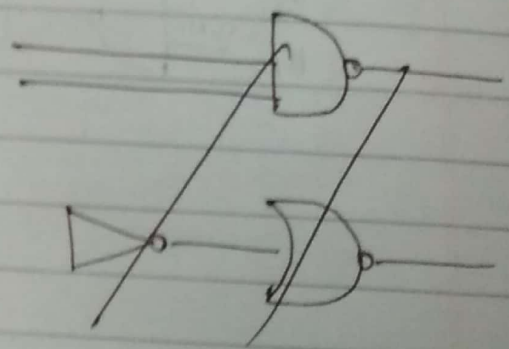


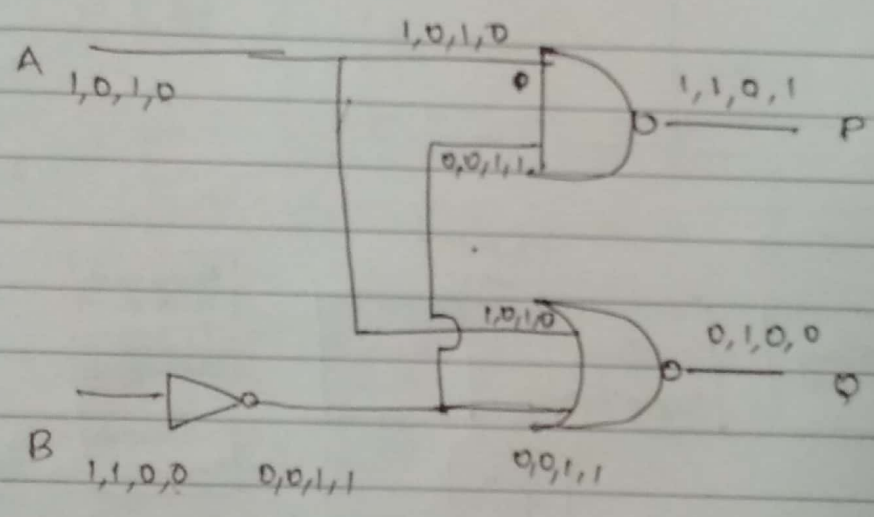
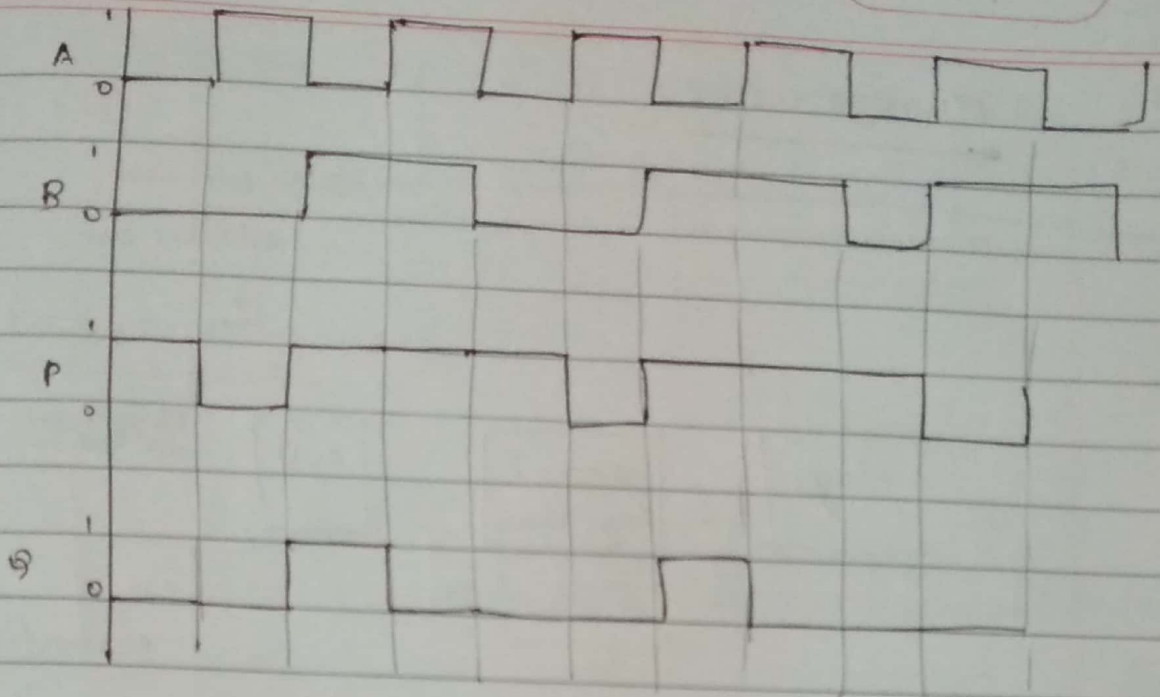
1. A.



2. A.

A	B	p	q
1	1	1	0
0	1	0	0
1	0	0	0
0	0	0	0



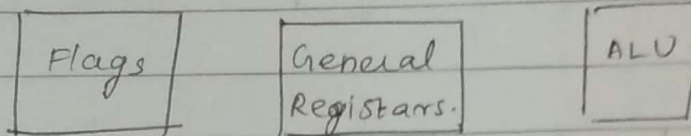


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Microprocessor

Shift
For what
how
why

data bus
Address bus

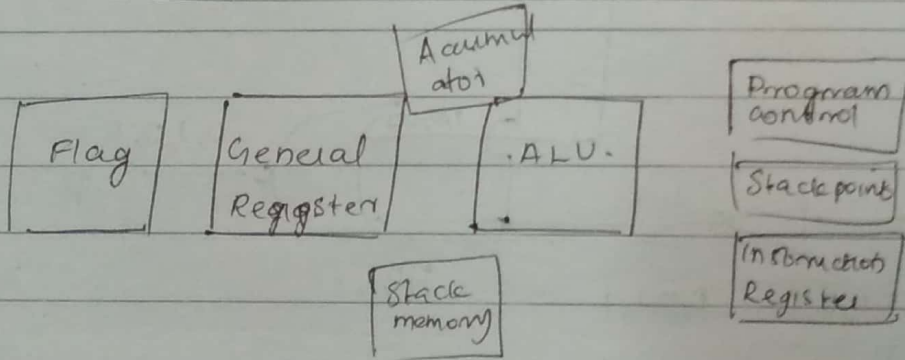


Program control

control bus

Data bus

Address bus



MAR

Control unit

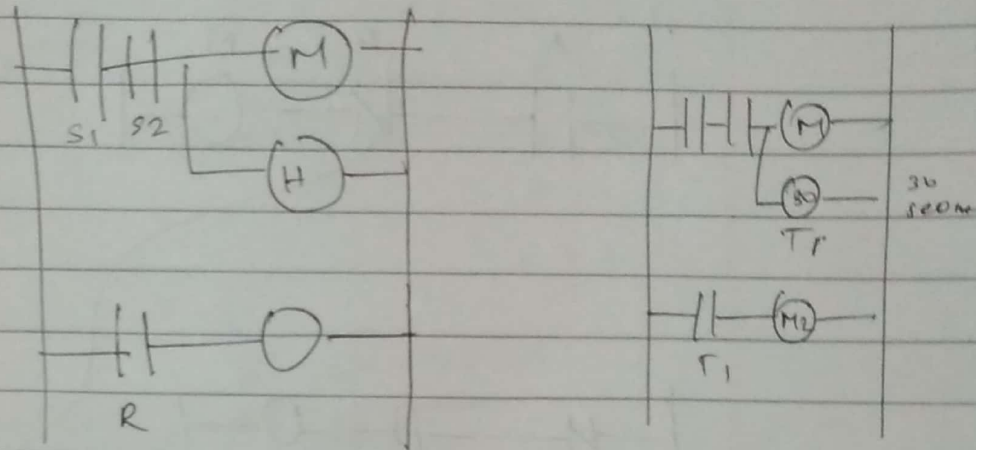
Control bus

Z → 0
N → 1 0
V → Over Flow.
C

0
1
1

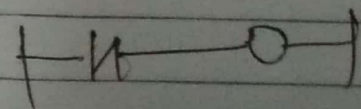
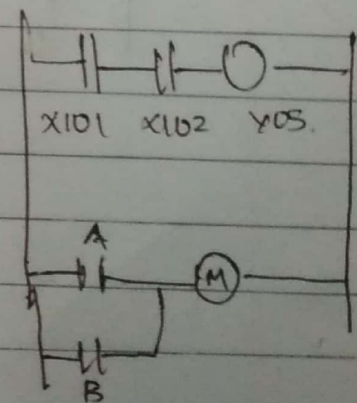
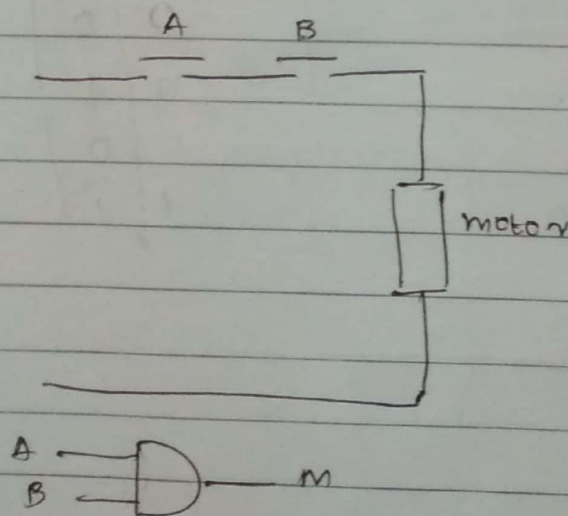
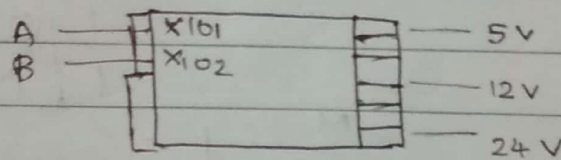
PROGRAMMABLE LOGIC CONTROLLER (PLC)

Ladder Diagram

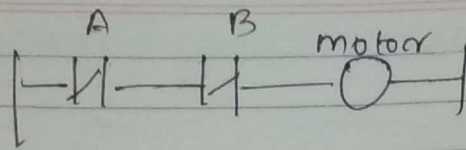


M2 starts after 30 sec

12/03/18

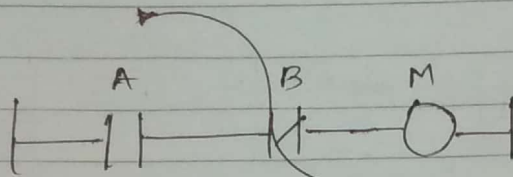


Stop switch
Normally closed. When

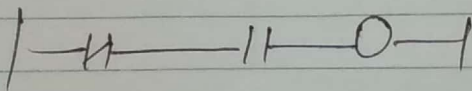


NOR Gate

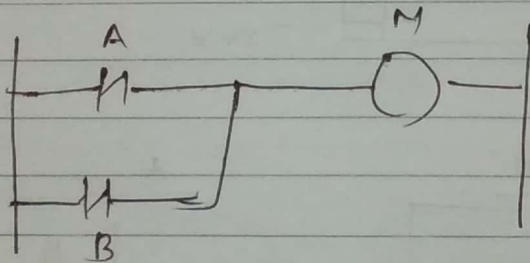
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0



A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0



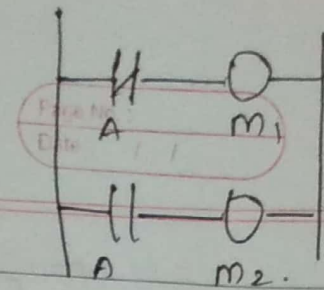
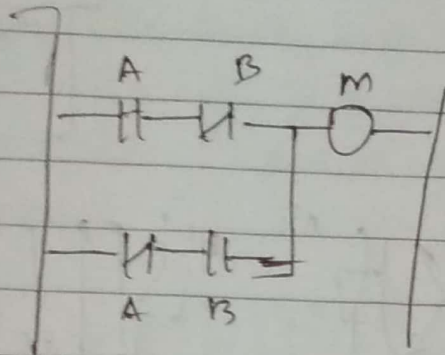
A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0



NAND

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

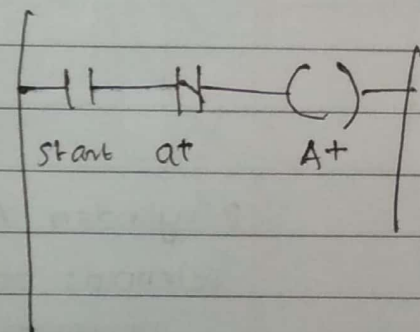
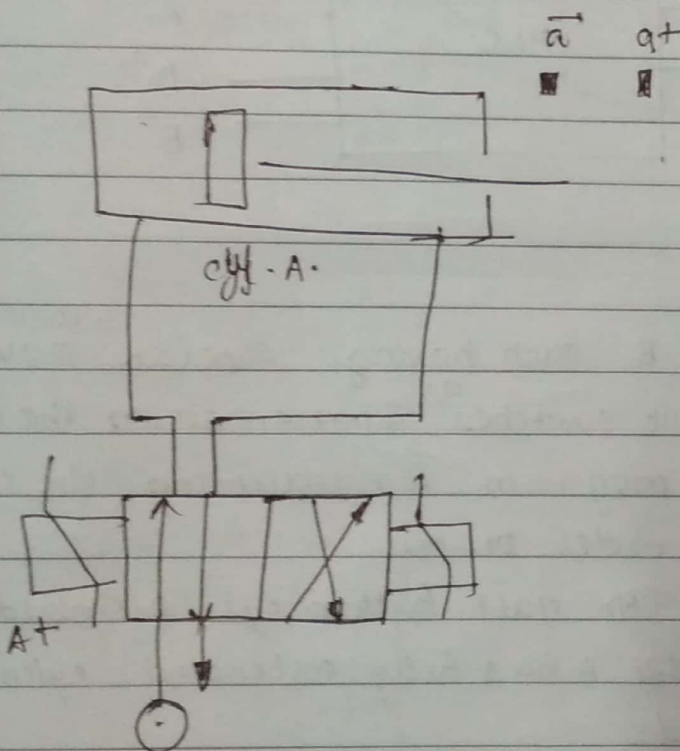
XOR Gate

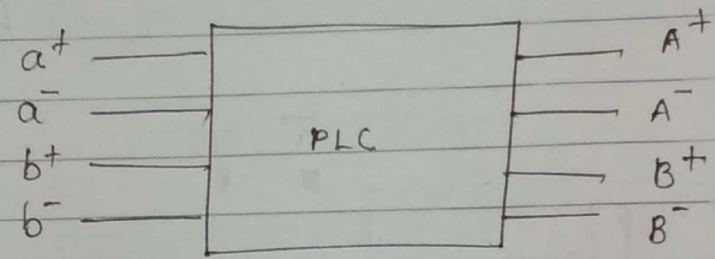
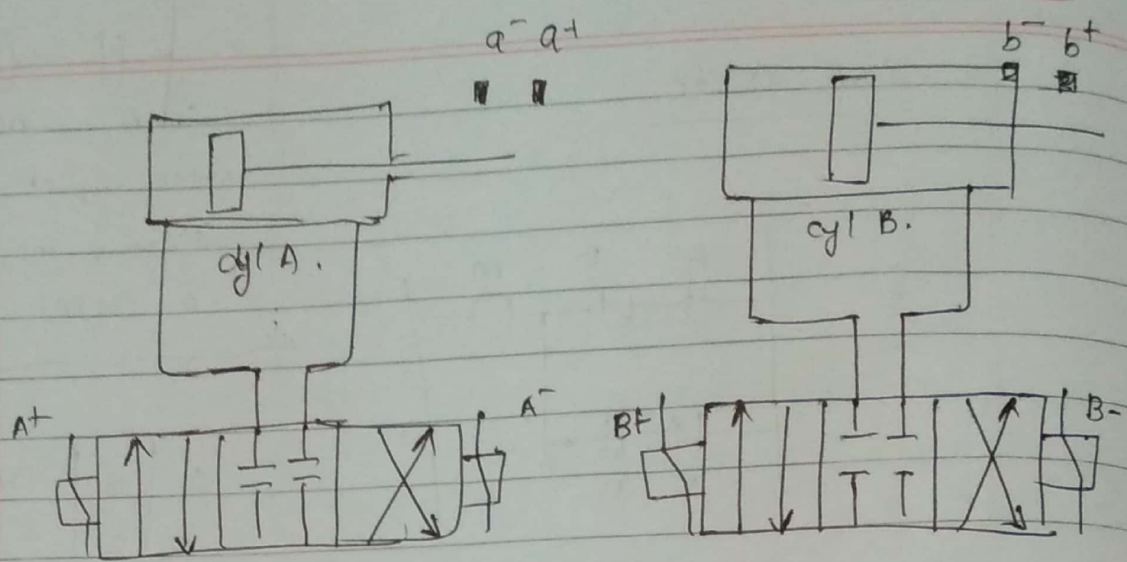


when input given A
opens.
closes, m1 stops.

A closes, m2 starts

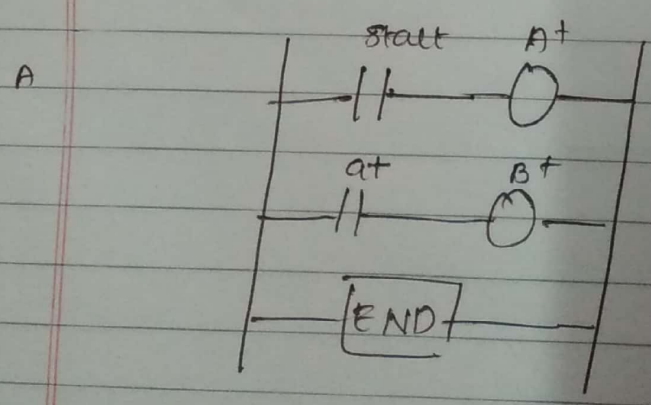
A	B	Y
0	0	0
1	0	1
0	1	1
1	1	0





2 cylinder A and B each having 3 position separate DCV with solenoids and limit switches represented in the figure. write the ladder program for sequencing the cylinder in the following order on page

- on pressing the start button cyl A should extend
- when cylinder A has fully extended cylinder B should extend

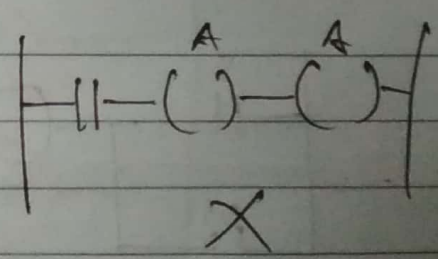
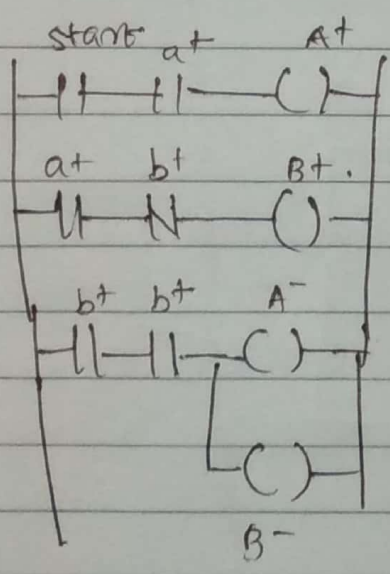
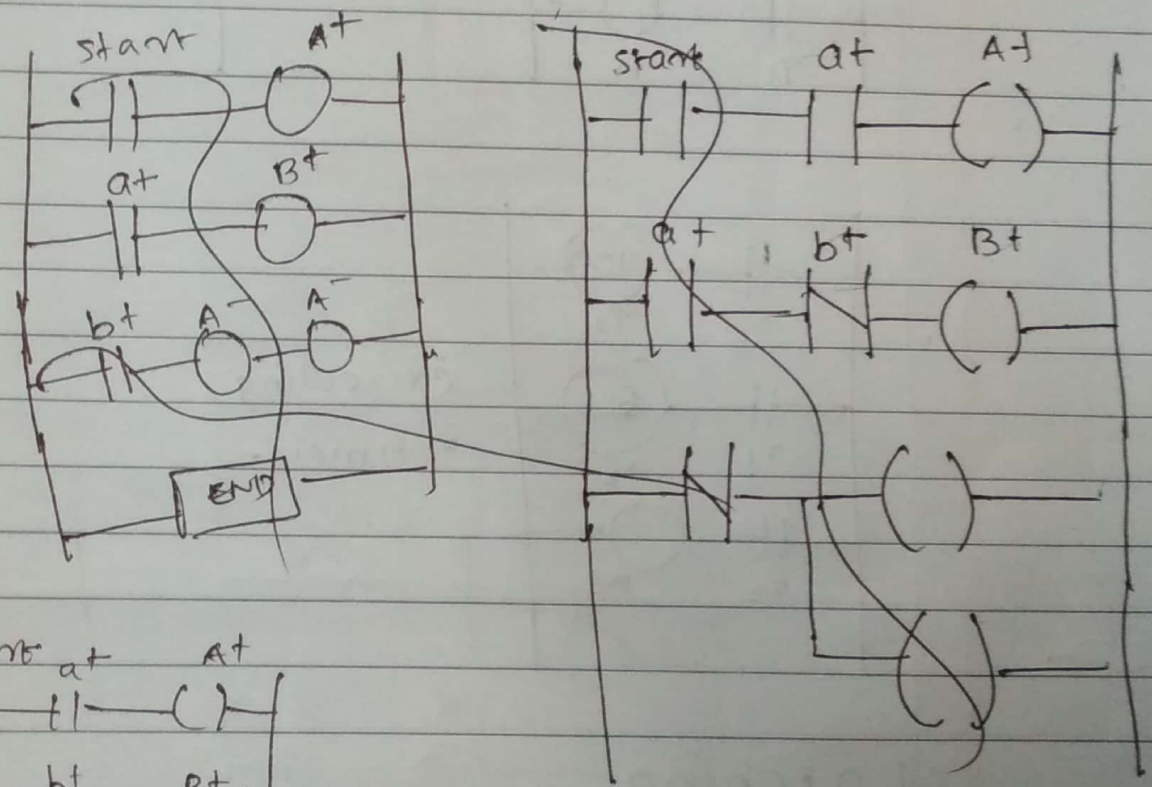


2. Write the ladder programme for obtaining the following sequences on

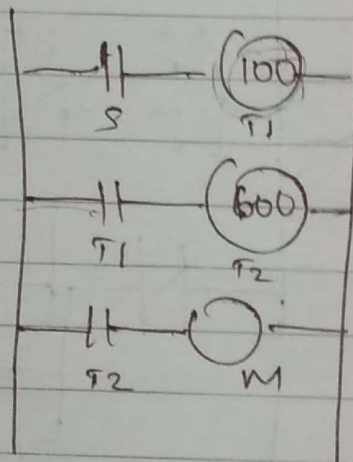
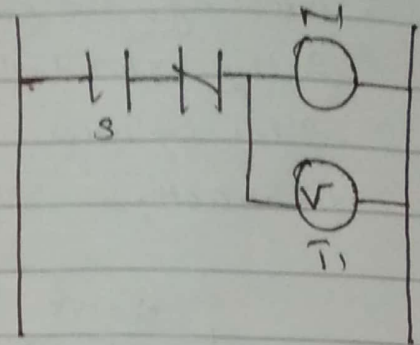
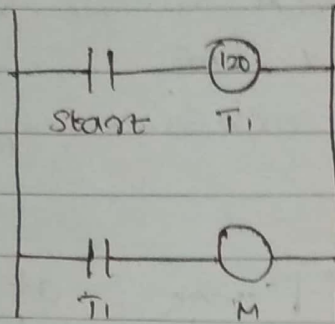
→ Pressing the start button cyl A extends, when cyl A is fully extended cyl B extends. AF

After cylinder A and B have extended fully both A & B retract.

A.

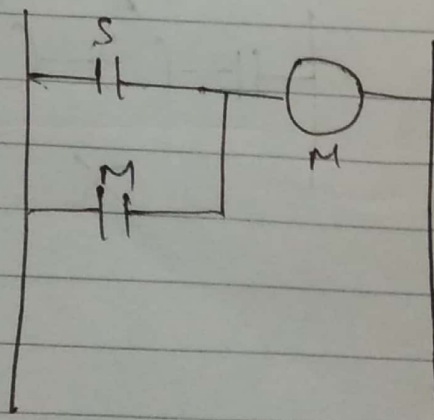


Timer



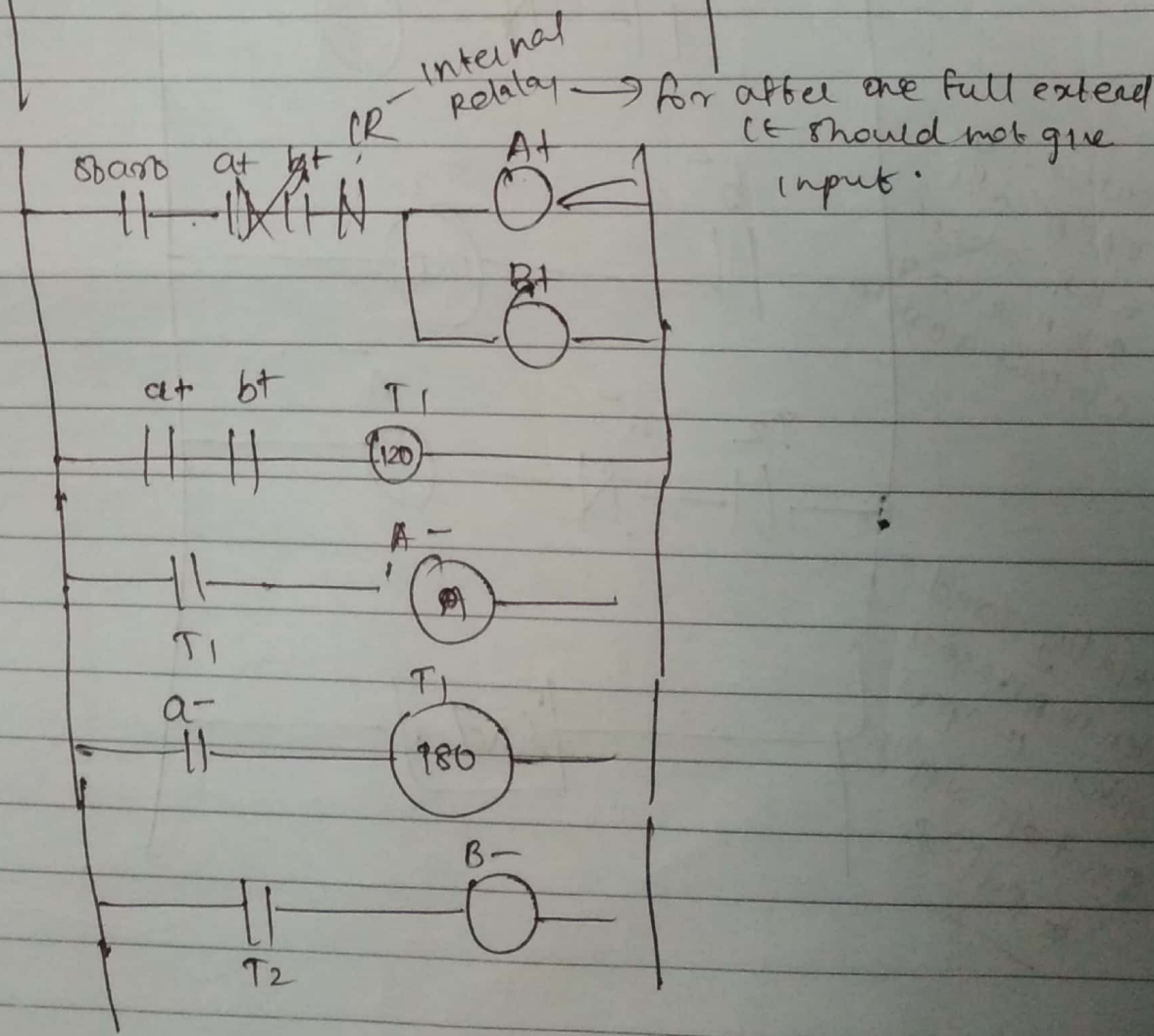
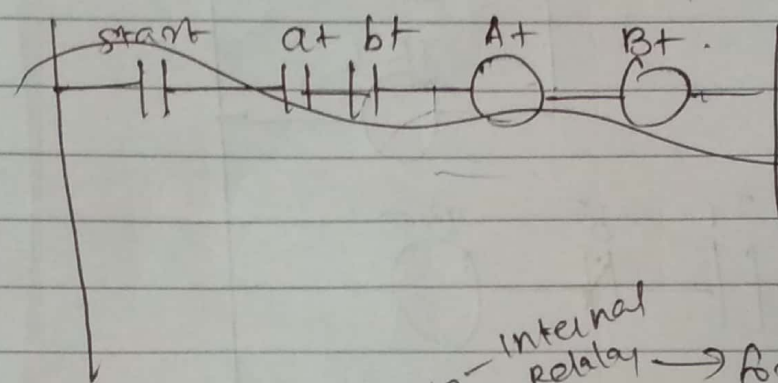
cooling
 9 times.

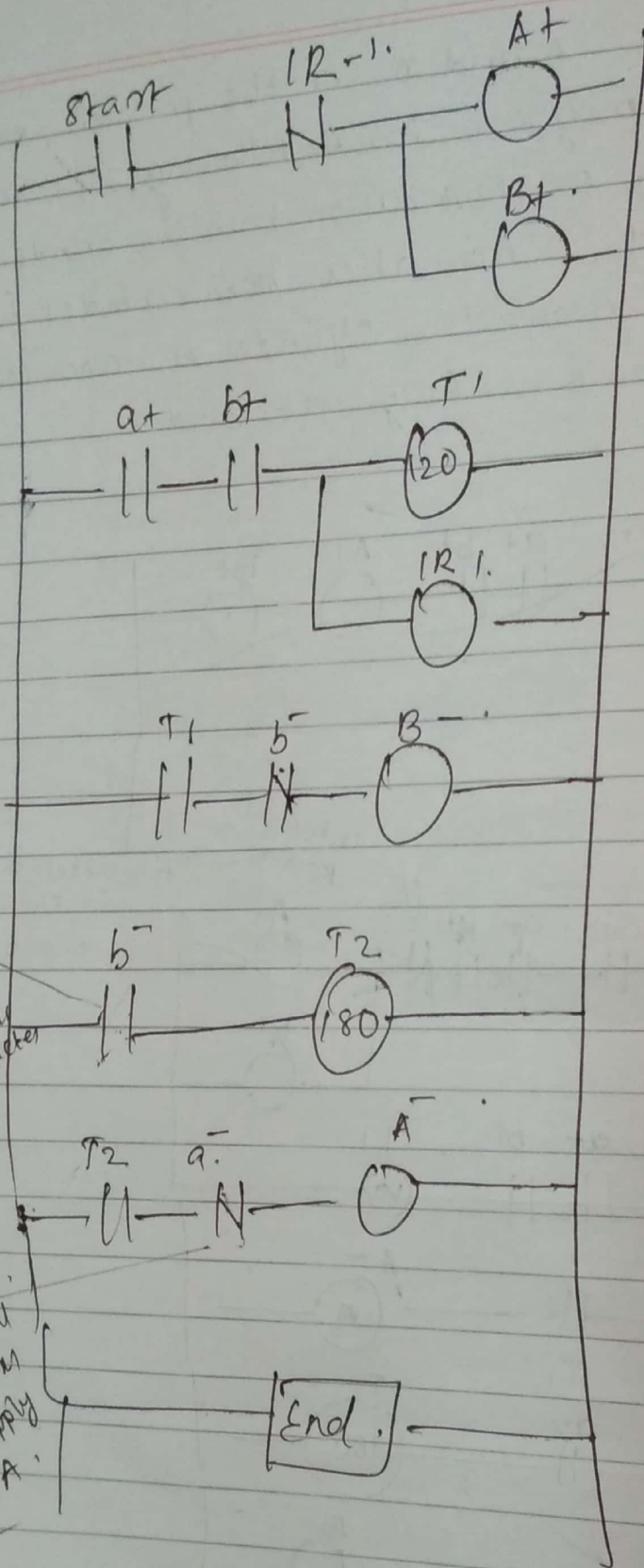
Latching



Q. For cylinders A and B of the previous case ^{write} ~~long~~ a ladder program for the following sequence on pressing the start button both the cylinders A and B should extend 2 min after both cylinders fully extended cylinder B retracts, cylinder A retracts 3 min after cylinder B has fully retracted.

B.





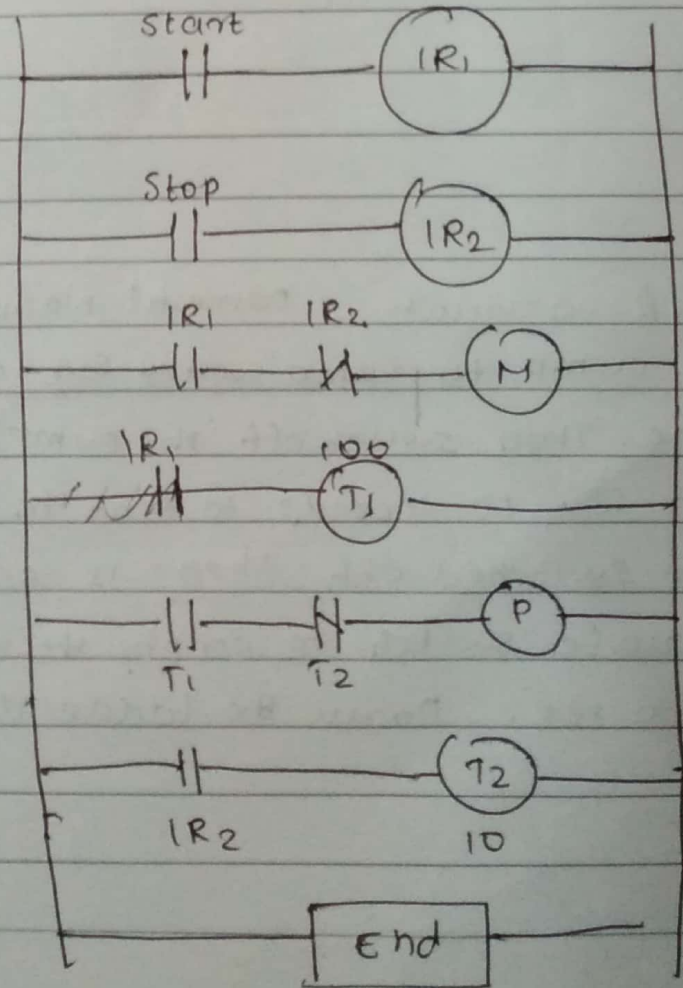
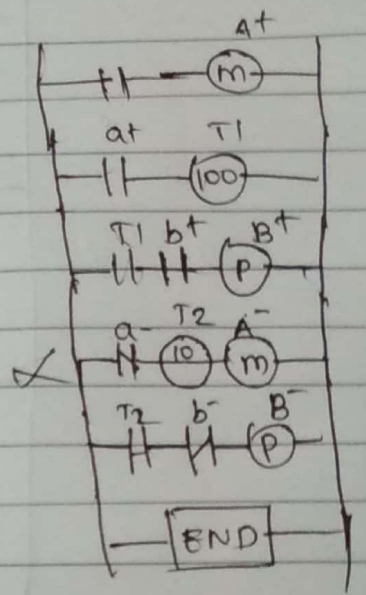
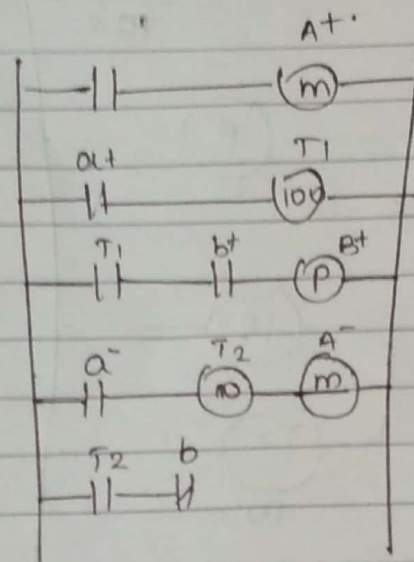
initially open coil
when signal given it closes
circuit completes

initially closed,
when input signal gives it opens
cut off supply to cyl A.

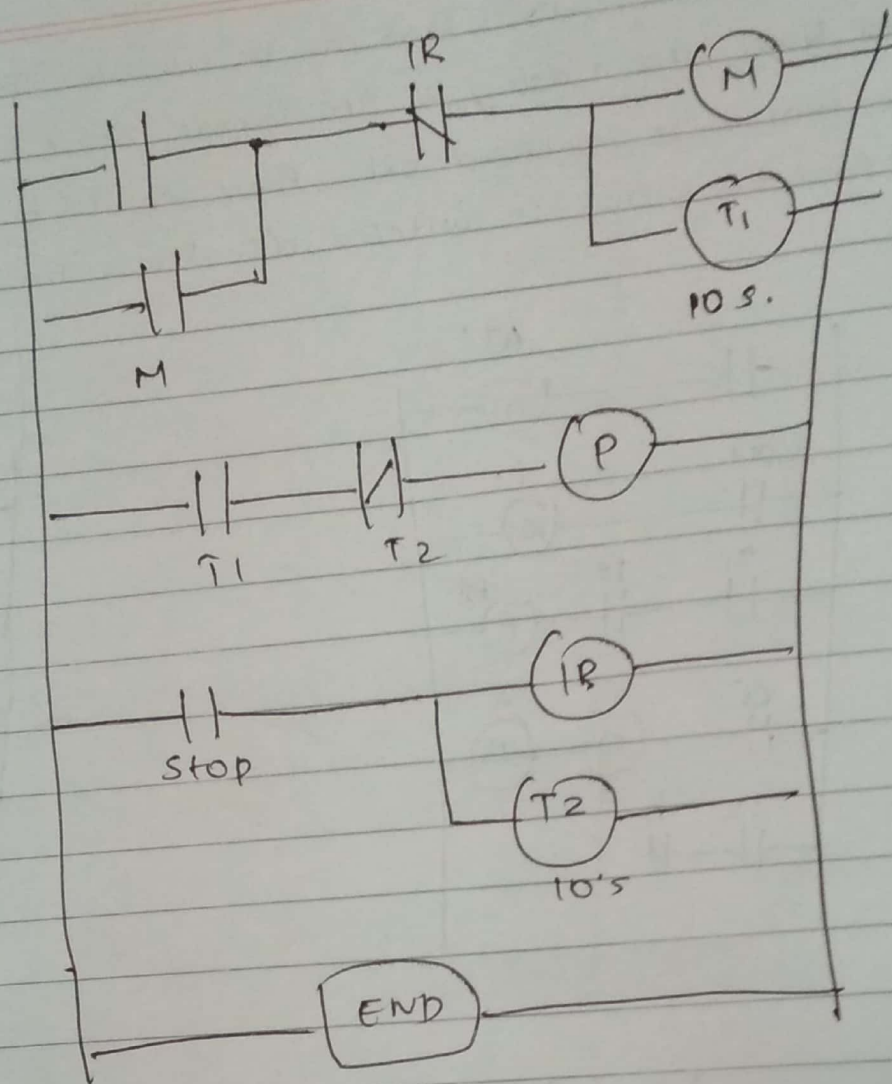
11/11/11

9. Design a circuit that can be used to start a motor and then after a delay of 100 seconds, start a pump. When the motor is switched off there should be a delay of 10 seconds before the pump is switched off. Draw the ladder diagram.

A.



Latching



- Q. Design a circuit for Domestic Washing machine to switch ON a pump to pump water for 100 sec, into the machine. Then switch off the pump and switch ON a heater for 50 seconds to heat the water. The heater is then switched off. There is another pump with a separate switch to empty the water from machine for 100 sec. Draw the ladder diagram.

Washing machine → Pump, heater.

